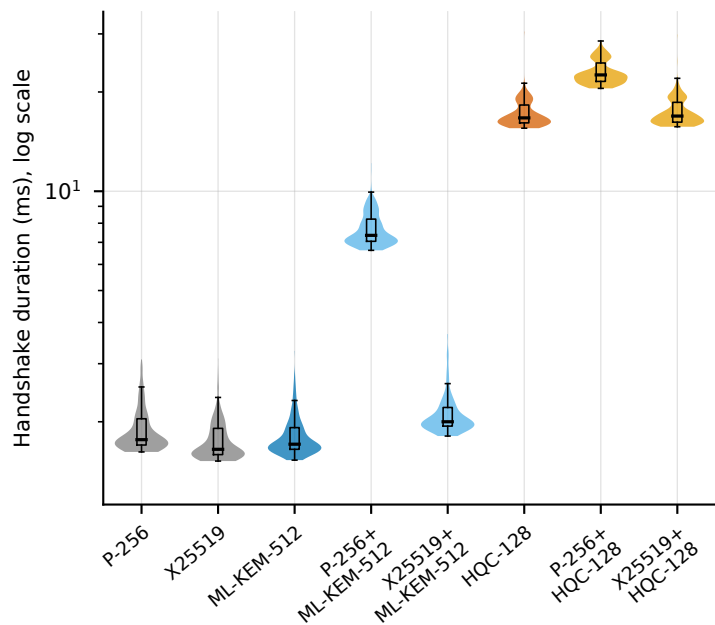
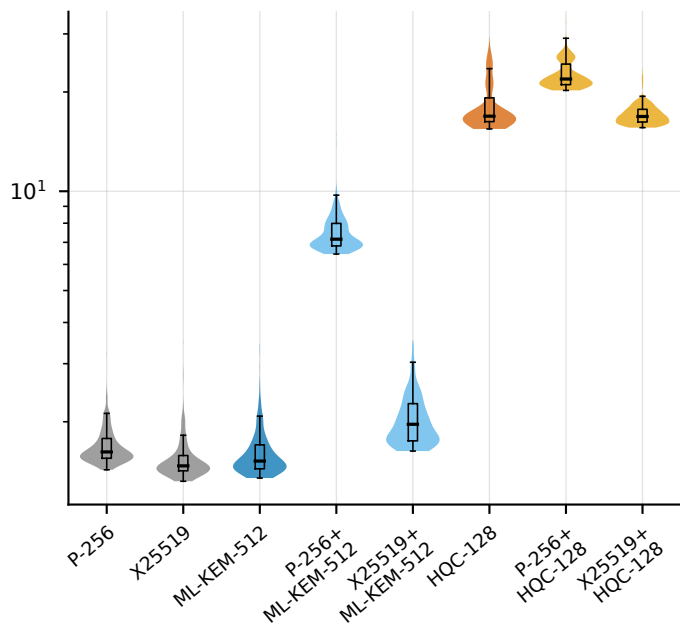


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	1.91	16.2	3.09	1.70	12.6	3.25
X25519	1.77	14.2	3.11	1.54	14.3	3.53
ML-KEM-512	1.83	14.0	3.28	1.62	16.5	3.43
P-256+ML-KEM-512	7.70	11.4	12.2	7.49	12.2	15.1
X25519+ML-KEM-512	2.12	14.5	3.69	2.06	18.2	3.53
HQC-128	17.2	8.8	30.5	18.2	16.8	29.3
P-256+HQC-128	23.0	8.1	28.6	22.7	9.7	39.5
X25519+HQC-128	17.4	9.2	29.9	17.1	6.8	23.5